

SIEVE LIMITS FOR THE ERDŐS–STRAUS EXCEPTIONAL SET: WHY $3/4$ IS SHARP FOR CONGRUENCE SIEVES

ANONYMOUS

ABSTRACT. Let $E(N)$ count the $n \leq N$ for which $4/n$ is not a sum of three positive unit fractions. The best internally proved bound in this programme is $E(N) \ll N \exp\{-c(\log N)^{3/4}\}$. We show that the exponent $3/4$ is a hard ceiling for a precisely defined class of congruence sieves. The class consists of nonnegative CRT majorants that are at least one on the whole avoider set of a family of forced congruence classes, whose final bound has the form $N \cdot \mathbb{E}\nu + \sum_i |a_i|$, and whose family moduli are at most $N^{O(1)}$. For families in which every modulus has a dominant prime factor, and, for the classes $\mathcal{R}(M)$, for families of gapped moduli, every such sieve saves at most $O((\log N)^{3/4})$. No power of $\log \log N$ can be gained. The majorant of the $3/4$ note lies in this class, and the large-sieve argument of the $2/3$ -loglog note is capped in the same way at its own exponent. We list exactly what the theorems exclude. For Selberg-type majorants we prove a sieve limit with no structural hypothesis, in terms of a noise-stability quantity, and a two-prime cap with twin moduli that is conditional on an equidistribution hypothesis. Every statement carries its status label. All proofs were checked internally only; nothing here has been externally refereed. This is not a proof of the Erdős–Straus conjecture.

1. INTRODUCTION AND STATUS

Call $n \geq 1$ *representable* if $4/n = 1/x_1 + 1/x_2 + 1/x_3$ with positive integers x_i , and let $E(N)$ count the non-representable $n \leq N$. The Erdős–Straus conjecture asserts that every $n \geq 2$ is representable. Vaughan [V] proved $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$ with Montgomery’s large sieve [M]. Two companion notes of this programme improve this. The first [LL] gains a factor $(\log \log N)^{1/3}$ in the exponent. The second [TQ] proves

$$E(N) \ll N \exp\{-c(\log N)^{3/4}\},$$

with label *internally proved; not externally refereed*. The $3/4$ note itself explains $3/4$ heuristically: the forced classes have cubic logarithmic mass, and a mass-driven optimisation gives $\theta = B/(B+1)$ with $B = 3$ [TQ, §10].

This note turns that heuristic into a theorem for a precise class of sieves, and records exactly where the theorem stops.

Status labels. Every statement carries one of the following labels.

- [PROVED]: proved here, with the proof given or precisely cited to an internal document; checked by internal hostile review only.
- [PROVED MOD X]: proved assuming the published external result X, which was checked against the source but not re-proved.
- [CONDITIONAL]: proved assuming a named open hypothesis.

Date: Research note; internally reviewed results only, not externally refereed.

- [EVIDENCE]: numerical or model evidence; never a proof.
- [CONJECTURE] / [OPEN]: not proved.

Apart from the classical facts of Section 2, each [PROVED] result below appears in one of four internal working documents, [ET], [EB], [TW], [TW2]. Each was hostile-reviewed; the reviews are [ETr, EBr, TWr, TW2r]. No prior source for the sieve-limit theorem (Theorem 4.5) was found; it may be folklore in spirit. Novelty is not claimed.

Results.

- (1) *The sieve-limit theorem* (Theorem 4.5, [PROVED]). For a prime-slice CRT system every nonnegative majorant of level λ of the avoider set saves at most $C \inf_{\alpha} [\alpha \lambda + \sum_{\ell} \bar{p}_{\ell} \ell^{-\alpha}] + O(\log^2 \lambda)$ beyond an admissible-set term. A sequential version allows shared large primes (Theorem 4.9, [PROVED]).
- (2) *The 3/4 cap* (Theorem 6.3). The forced classes have cubic profile $\sum \bar{p}_{\ell} \ell^{-\alpha} \ll \alpha^{-3}$ (Section 5). Hence every CRT-majorant architecture (Definition 3.3: $\nu \geq 0$ on $\mathbb{Z}$, $\nu \geq 1$ on the whole avoider set, final bound $N\mathbb{E}\nu + \sum |a_i|$, family primes $\leq N^{O(1)}$) saves at most $O((\log N)^{3/4})$, for families whose moduli all have a dominant prime ([PROVED]; Case-A classes mod Elsholtz–Tao Prop. 1.4) and for (η, B) -gapped $\mathcal{R}(M)$ -families ([PROVED]).
- (3) *The campaign sieves are in the class* (Section 8, [PROVED]). The majorant of the 3/4 note is in the class, so 3/4 is sharp for its architecture; the large sieve of the 2/3-loglog note is capped at its own exponent $\frac{2}{3}$ with the $(\log \log N)^{1/3}$ factor.
- (4) *Exclusions* (Section 9). A precise list of what lies outside, including unresolved twin moduli, which carry cubic supply (Lemma 9.1, [PROVED]).
- (5) *Selberg-type majorants* (Section 10). A Λ^2 sieve limit for arbitrary systems in terms of noise stability (Theorem 10.1, [PROVED]); a two-prime Λ^2 cap with twin moduli, [CONDITIONAL] on the open equidistribution hypothesis ($H_O^{\neq}$) (Corollary 10.8).
- (6) *Open problems* (Section 11): H_{MS} , the k -ary comparison inequality (Conjecture 11.2), sparse noise stability, and ($H_O^{\neq}$).

None of this bounds $E(N)$ from below or says anything about methods outside the class; in particular it is not evidence against the conjecture.

2. FORCED CLASSES

A residue class $b \pmod{G}$ is *forced* if every positive integer $n \equiv b \pmod{G}$ is representable. We use three families of forced classes. For an integer $D \geq 1$ put $g(D) = \prod_p p^{\lceil v_p(D)/2 \rceil}$, so that $g(D) \mid D \mid g(D)^2$ and, for every integer x , $D \mid x^2 \iff g(D) \mid x$.

Lemma 2.1 (multiplier identity; [LL, Lemma 2.1]; [PROVED]). *Let $M \equiv 3 \pmod{4}$, $A = (M+1)/4$, and $A = uvw$ with positive integers u, v, w . If $n \geq 1$ and $nv \equiv -u \pmod{M}$, then $s = (nv+u)/M$ is a positive integer and $4/n = 1/(suw) + 1/(nsvw) + 1/(nuvw)$.*

Proof. On the common denominator $nsuvw$ the numerators sum to $nv + u + s = s(M+1) = 4suvw$. $\square$

Lemma 2.2 (the classes $\mathcal{R}(M)$; [N, Lemma 18.1]; [PROVED]). *For $M \equiv 3 \pmod{4}$ let $\mathcal{R}(M)$ be the set of classes $-uv^{-1} \pmod{M}$ over all factorisations $A = uvw$. Then*

$$\mathcal{R}(M) = \{-4D \pmod{M} : D \mid A^2\}, \quad |\mathcal{R}(M)| \leq \tau(A^2),$$

and every class of $\mathcal{R}(M)$ is forced.

Proof. Since $4A \equiv 1 \pmod{M}$, a factorisation $A = uvw$ gives $-uv^{-1} \equiv -u^2wA^{-1} \equiv -4u^2w \pmod{M}$ with $u^2w \mid A^2$. Conversely, if $A = \prod p^{e_p}$ and $D = \prod p^{d_p}$ with $d_p \leq 2e_p$, put $v_p(u) = \lfloor d_p/2 \rfloor$, $v_p(w) = d_p - 2\lfloor d_p/2 \rfloor$ and $v_p(v) = e_p - v_p(u) - v_p(w) \geq 0$; then $uvw = A$ and $u^2w = D$. Forcedness is Lemma 2.1. $\square$

Lemma 2.3 ((a, D) -classes; [ET, Lemma 3.2]; [PROVED]). *For all integers $a, D \geq 1$ the class $n \equiv -(4D + a) \pmod{4a g(D)}$ is forced.*

Proof. Write $n + 4D + a = 4a g(D) j$ with $j \geq 1$. Then $M := (n + 4D)/a = 4g(D)j - 1$ is $\equiv 3 \pmod{4}$ and $(M + 1)/4 = g(D)j$, so $D \mid g(D)^2 \mid ((M + 1)/4)^2$. Since $n \equiv -4D \pmod{M}$, n lies in a class of $\mathcal{R}(M)$, and Lemma 2.2 applies. $\square$

These classes encode exactly the Case-B witnesses of the classical parametrisation [N, Thm 3.1(B)] (see [ET, Lemma 3.2]).

Lemma 2.4 (Case-A classes; [ET, §3]; [PROVED]). *For $d \geq 1$ and $m \mid 4d + 1$ (so m is odd and $(m, 4g(d)) = 1$), the class $n \equiv -m^{-1} \pmod{4g(d)}$ is forced.*

Proof. Put $z_0 = (nm + 1)/4$, an integer. Then $g(d) \mid z_0$, so $d \mid z_0^2$; and $m \mid 4d + 1$ gives $m \mid 4d + 4z_0$, so $m \mid d + z_0$. With $x = (z_0 + d)/m$ and $y = (z_0 + z_0^2/d)/m = z_0(d + z_0)/(dm)$, both integers since $(m, d) = 1$, one has $1/x + 1/y = m/z_0$, hence $1/x + 1/y + 1/(nz_0) = (nm + 1)/(nz_0) = 4/n$. $\square$

We write *Case-B classes* for the classes of $\mathcal{R}(M)$ and the (a, D) -classes, and *Case-A classes* for those of Lemma 2.4. Two facts are used repeatedly.

Lemma 2.5 ($n = 1$ is never covered; [PROVED]). *No forced class contains $n = 1$.*

Proof. $4/1 > 3 \geq 1/x_1 + 1/x_2 + 1/x_3$. $\square$

Lemma 2.6 (Mordell's obstruction; classical; [TW, Lemma 1.1]; [PROVED]). *Let $M \equiv 3 \pmod{4}$, $A = (M + 1)/4$ and $D \mid A^2$. Then $(D, M) = 1$ and the Jacobi symbol satisfies $(\frac{-4D}{M}) = -1$. Consequently, if n is a nonzero quadratic residue modulo every prime factor of M , then n lies in no class of $\mathcal{R}(M)$.*

Proof. $(A, M) = (A, 4A - 1) = 1$. Write $D = sr^2$ with s squarefree; then $D \mid A^2$ gives $sr \mid A$, so $M \equiv -1 \pmod{4s}$. Now $(\frac{-4D}{M}) = (\frac{-1}{M})(\frac{s}{M}) = -(\frac{s}{M})$. If s is odd, reciprocity and $(\frac{M}{s}) = (\frac{-1}{s})$ give $(\frac{s}{M}) = 1$, since $(M - 1)/2$ is odd. If $s = 2s'$, then $8 \mid M + 1$, so $(\frac{2}{M}) = 1$, and $(\frac{s'}{M}) = 1$ as before. For the consequence, $(\frac{n}{M}) = \prod_{p \mid M} (\frac{n}{p})^e = 1$. $\square$

3. CONGRUENCE SIEVES: THE ARCHITECTURE CLASS

3.1. CRT systems and majorants.

Definition 3.1 (prime-slice CRT system [ET, §1]). The data are a modulus Q_0 ; a finite set $\mathcal{P}$ of primes not dividing Q_0 (the *slice primes*); a set $R \subseteq \mathbb{Z}/Q_0$ of admissible small residues; and, for each $c \in R$ and $\ell \in \mathcal{P}$, a set $F_\ell(c) \subseteq \mathbb{Z}/\ell$ of forbidden classes. The *avoider set* is

$$\mathcal{A} = \{n : n \bmod Q_0 \in R, n \bmod \ell \notin F_\ell(n \bmod Q_0) \text{ for all } \ell \in \mathcal{P}\}.$$

Put $p_\ell(c) = |F_\ell(c)|/\ell$ and $\bar{p}_\ell = |R|^{-1} \sum_{c \in R} p_\ell(c)$.

A family of conditions $n \equiv b \pmod{q_0 \ell}$ with $q_0 \mid Q_0$ and $\ell \in \mathcal{P}$ induces such a system with $F_\ell(c) = \{b \bmod \ell : b \equiv c \pmod{q_0}\}$. A typical R is a *selector* $R = \{c : (c, P) = 1\}$ with $P \mid Q_0$.

Definition 3.2 (majorant of level λ). A *majorant* of $\mathcal{A}$ is a finite real combination

$$\nu(n) = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$$

with $\nu \geq 0$ on all of $\mathbb{Z}$ and $\nu \geq 1$ on $\mathcal{A}$. It has *level* λ if $\sum_{\ell \in \mathcal{P}, \ell \mid d_i} \log \ell \leq \lambda$ for every i . Its *mean* is $\mathbb{E}\nu = \sum_i a_i/d_i$, the average over a common period, and its *saving* is $\log(1/\mathbb{E}\nu)$.

The level charges only the slice primes; the small part of d_i is free. This is generous to the sieve. For every N ,

$$\sum_{n \leq N} \nu(n) = N \cdot \mathbb{E}\nu + O\left(\sum_i |a_i|\right), \quad (1)$$

since each class meets $[1, N]$ in $N/d_i + O(1)$ points.

3.2. The architecture class.

Definition 3.3 (CRT-majorant architecture). A *CRT-majorant architecture* for a family $\mathfrak{F}$ of forced classes bounds $\#(\mathcal{A} \cap [1, N])$, and hence the exceptional integers in $[1, N]$ that satisfy the admissibility condition $n \bmod Q_0 \in R$, as follows.

- (A1) ν is a majorant of the avoider set $\mathcal{A}$ of $\mathfrak{F}$ in the sense of Definition 3.2: $\nu \geq 0$ on $\mathbb{Z}$ and $\nu \geq 1$ on the *whole* avoider set, not only on exceptional primes.
- (A2) The final bound is $N \cdot \mathbb{E}\nu + T$ with $T = \sum_i |a_i|$, and it is non-trivial: $T < N$.
- (A3) The slice primes of the family are at most N^A for a fixed A .

Every sieve majorant in use satisfies (A1): Bonferroni truncations $Q_r(H) = \binom{H-1}{r}$ and Selberg squares. Both campaign notes satisfy (A1)–(A3) (Section 8).

Lemma 3.4 (coefficient budget implies level; [ET, Lemma 2.9], from [ETr, item 3]; [PROVED]). *Take a prime-slice system, or the sequential setting of Section 4.4. Let ν be a majorant of $\mathcal{A}$ with $T = \sum_i |a_i|$, and assume that every slice prime satisfies $\log \ell \leq \Lambda_0$. Then for every λ there is a majorant ν' of level at most λ with $\nu' \geq \nu$ pointwise and $\mathbb{E}\nu' \leq \mathbb{E}\nu + Te^{\Lambda_0 - \lambda}$. In particular $\lambda = \Lambda_0 + \log T + \log(1/\mathbb{E}\nu)$ gives $\mathbb{E}\nu' \leq 2\mathbb{E}\nu$.*

Proof. Keep every term of level at most λ . Drop each term of level $> \lambda$ with $a_i < 0$; this raises ν pointwise and raises the mean by $|a_i|/d_i \leq |a_i|e^{-\lambda}$. For a term of level $> \lambda$ with $a_i > 0$, list its slice primes in increasing order and keep the longest prefix of level at most λ . The next prime has $\log \ell \leq \Lambda_0$, so the prefix has level $> \lambda - \Lambda_0$. Replace d_i by the divisor d'_i formed by the non-slice part of d_i and the kept prime powers. Then $\mathbf{1}[n \equiv b_i \pmod{d'_i}] \geq \mathbf{1}[n \equiv b_i \pmod{d_i}]$, and the mean rises

by at most $a_i/d'_i \leq a_i e^{\Lambda_0 - \lambda}$. So $\nu' \geq \nu$, ν' is a majorant, and summing gives the bound. $\square$

Corollary 3.5 ([PROVED]). *Under (A2)–(A3), the majorant ν can be replaced by a majorant of level $\lambda \leq (A+1) \log N + s$ and mean at most $2\mathbb{E}\nu$, where $s = \log(1/\mathbb{E}\nu)$. Hence, if every majorant of level λ has saving at most $C\lambda^{3/4} + C'$, then $s \ll_{A,C,C'} (\log N)^{3/4}$.*

Proof. Lemma 3.4 with $\Lambda_0 = A \log N$ and $\log T < \log N$ gives ν' of level $\lambda \leq (A+1) \log N + s$ and saving at least $s - \log 2$. So $s \leq \log 2 + C((A+1) \log N + s)^{3/4} + C'$, which forces $s \ll_{A,C,C'} (\log N)^{3/4}$. $\square$

Remark 3.6 (positivity on all of $\mathbb{Z}$). Condition (A1) cannot be weakened to positivity on $[1, N]$. Restricted to $[1, N]$, the class $\mathbf{1}[n \equiv b \pmod{N}]$ is the point mass at b , so the span of level- $\log N$ class indicators contains every function on $[1, N]$. The relaxed linear programme is then the exact count, and evaluating it presupposes knowing the avoider set [ET, §2.4]. Such methods are not sieves in the present sense.

4. THE SIEVE-LIMIT THEOREM

All results of this section are from [ET, §2]; the review [ETr, items 1–2] found them sound and checked every intermediate inequality against exact linear programmes on small systems. Write $C_4 = 8e^5$.

4.1. Two one-dimensional lemmas.

Lemma 4.1 (binomial mass; [PROVED]). *Let $\psi = \text{Bin}(z, q)$ with $0 < q \leq 1/4$ and $\mu' = zq$. For every integer $1 \leq y \leq z-1$, $\psi(y) \geq (8y)^{-1/2} \exp\{-\frac{4}{3}(y - \mu')^2/\mu'\}$.*

Proof. By [MS, Ch. 10, Lemma 7], $\binom{z}{y} \geq (8y(1 - y/z))^{-1/2} e^{zH(y/z)}$, so $\psi(y) \geq (8y)^{-1/2} e^{-zD(y/z||q)}$. The bound $D(a||q) \leq (a - q)^2/(q(1 - q))$ gives $zD \leq (y - \mu')^2/(\mu'(1 - q)) \leq \frac{4}{3}(y - \mu')^2/\mu'$. $\square$

Lemma 4.2 (node lemma; [PROVED]). *Keep ψ, q, μ' as in Lemma 4.1. For a set Y of $k+1$ distinct integers in $[0, z]$ let $B(Y) = \max_{y \in Y} |\ell_y^Y(0)|/\psi(y)$, where ℓ_y^Y is the Lagrange basis polynomial of Y . Then every polynomial P of degree at most k with $P \geq 0$ on Y satisfies $|P(0)| \leq B(Y) \sum_{y \in Y} \psi(y)P(y)$. Let $B(k) = \min_Y B(Y)$ and $C_1 = 16e^6$. Then*

- (a) $\log B(0) \leq \frac{1}{2} \log(8\mu' + 8) + 2$;
- (b) $\log B(k) \leq 4 \log(4/3)\mu' \leq 1.151\mu'$ for every $k \leq z$;
- (c) if $\mu' \geq 64$ and $1 \leq k \leq \mu'/16$, then $\log B(k) \leq \frac{k}{2} \log(C_1\mu'/k) + \frac{1}{2} \log(16\mu')$.

Consequently, for all $\alpha, s > 0$ and $0 \leq k \leq z$,

$$\log B(k) \leq 19\alpha ks + C_4\mu' e^{-2\alpha s} + \frac{1}{2} \log(16\mu' + 16) + 74. \quad (2)$$

Proof. The inequality for P is Lagrange interpolation $P(0) = \sum_y \ell_y(0)P(y)$ with $P \geq 0$ on Y . (a) If $\mu' \geq 1$ then $z \geq 4$; take the single node $y = \lceil \mu' \rceil \leq z-1$ and apply Lemma 4.1 with $|y - \mu'| \leq 1$. If $\mu' < 1$ take the node 0: $\psi(0) = (1 - q)^z \geq e^{-1.151\mu'}$. (b) Take $Y = \{0, \dots, k\}$; then $\ell_y(0) = \delta_{y0}$ and $B = 1/\psi(0) \leq e^{1.151\mu'}$. (c) Take nodes $y_i = a + ih$, $0 \leq i \leq k$, with $W = \lceil \sqrt{k\mu'}/2 \rceil$, $h = \lceil 2W/k \rceil$, $a = \lceil \mu' \rceil - W$. Since $k \leq \mu'/16$ and $\mu' \geq 64$, every node lies in $[\mu'/2, 2\mu'] \subseteq [1, z-1]$ and $|y_i - \mu'| \leq \sqrt{k\mu'}$, so Lemma 4.1 gives $\psi(y_i) \geq (16\mu')^{-1/2} e^{-4k/3}$. Next $|\ell_i(0)| = \prod_{j \neq i} y_j/(h^k i!(k-i)!)$,

and $\prod_{j \neq i} y_j \leq (2\mu')^k$, $k!/(i!(k-i)!) \leq 2^k$, $k! \geq (k/e)^k$ and $h \geq \sqrt{\mu'/k}$ give $|\ell_i(0)| \leq (4e\sqrt{\mu'/k})^k$. Multiplying gives (c), since $16e^{2+8/3} \leq C_1$.

For (2): when (c) applies, use $\frac{k}{2} \log(C_1\mu'/k) \leq \alpha ks + (C_1\mu'/2e)e^{-2\alpha s}$ (the maximum over real k of the difference), and $C_1/(2e) = C_4$. Otherwise use (a) or (b): if $\mu' < 64$ then $1.151\mu' \leq 74$; if $k > \mu'/16$ and $\alpha s \geq 1$ then $1.151\mu' \leq 18.5k \leq 19\alpha ks$; if $\alpha s < 1$ then $1.151\mu' \leq 8.6\mu'e^{-2\alpha s}$. $\square$

Lemma 4.3 (combination technique; standard; [PROVED]). *Let $\Lambda \subset \mathbb{Z}_{\geq 0}^G$ be a lower set and P_Λ the span of the monomials x^a , $a \in \Lambda$. For each coordinate g and degree k let $I_k^{(g)}$ be Lagrange interpolation at any $k+1$ distinct nodes, and put $c_j = \sum_{e \in \{0,1\}^G} (-1)^{|e|} \prod_{j+e \in \Lambda} (-1)^{|e|}$. Then for $Q \in P_\Lambda$, $Q = \sum_{j \in \Lambda} c_j (\bigotimes_g I_{j_g}^{(g)})Q$, and $|c_j| \leq 2^G$.*

Proof. Put $\Delta_k = I_k - I_{k-1}$ with $I_{-1} = 0$; expanding $\sum_{j \in \Lambda} \bigotimes_g \Delta_{j_g}$ gives the coefficients c_j . For $a \in \Lambda$, $\Delta_{j_g} x_g^{a_g} = 0$ when $j_g > a_g$, and the box $\{j \leq a\}$ lies in Λ , so the sum applied to x^a is $\prod_g \sum_{j_g \leq a_g} \Delta_{j_g} x_g^{a_g} = x^a$. $\square$

4.2. The Boolean proposition. Let $x_i \sim \text{Bern}(p_i)$, $i \in I$, be independent, with weights $s_i \geq s_* > 0$. A function f on $\{0,1\}^I$ is λ -level if $f = \sum_T f_T$ where f_T depends only on x_T and $\sum_{i \in T} s_i \leq \lambda$.

Proposition 4.4 ([ET, Prop. 2.4]; [PROVED]). *Assume $p_i \leq 1/4$ whenever $s_i \leq \lambda$, and $p_i < 1$ for all i . Let f be λ -level with $f \geq 0$ and $f(0) \geq 1$, and let $\lambda \geq s_*$. Put $G = \lceil \log_2(\lambda/s_*) \rceil + 1$ and $\mu = \sum_{s_i \leq \lambda} p_i$. Then for every $\alpha > 0$,*

$$\log \frac{1}{\mathbb{E}f} \leq 19\alpha\lambda + C_4 \sum_{s_i \leq \lambda} p_i e^{-\alpha s_i} + G(75 + \log(2 + \lambda/s_*)) + \frac{G}{2} \log(16\mu + 16). \quad (3)$$

Proof. Step 0. No f_T contains an i with $s_i > \lambda$, so f ignores those coordinates; discard them, coordinates with $p_i = 0$, and empty bands.

Step 1 (bands). Put $B_g = \{i : 2^g s_* \leq s_i < 2^{g+1} s_*\}$ for $0 \leq g < G$, $s_g = 2^g s_*$ and $q_g = \max_{i \in B_g} p_i \leq 1/4$.

Step 2 (thinning). Take independent $u_i \sim \text{Bern}(q_g)$ and $w_i \sim \text{Bern}(p_i/q_g)$; then $u_i w_i$ has the law of x_i . Fix w , let $Z_g = \{i \in B_g : w_i = 1\}$ and $z_g = |Z_g|$. Then $f_w(u) = f(u \circ w)$ depends only on u_Z , is λ -level and nonnegative, and $f_w(0) \geq 1$.

Step 3 (symmetrisation). Average f_w over $\prod_g \text{Sym}(Z_g)$, which preserves the law of u . Each monomial u^S of a term f_T has multidegree $(|S \cap Z_g|)_g$ in the lower set $\Lambda = \{j : \sum_g j_g s_g \leq \lambda\}$, and symmetrises to $\prod_g \binom{K_g}{|S_g|} / \binom{z_g}{|S_g|}$ with $K_g = |u_{Z_g}|$. Hence $\mathbb{E}_u f_w = \mathbb{E}Q(K)$ where $Q \in P_\Lambda$, the $K_g \sim \text{Bin}(z_g, q_g)$ are independent, $Q \geq 0$ on $\prod_g \{0, \dots, z_g\}$ and $Q(0) \geq 1$. Reducing modulo $\prod_{i=0}^{z_g} (K_g - i)$ we may take $Q \in P_{\Lambda'}$, $\Lambda' = \Lambda \cap \prod_g [0, z_g]$.

Step 4 (interpolation). Choose the node sets of Lemma 4.2 inside $\{0, \dots, z_g\}$ and apply Lemma 4.3 at 0. All grid points are support points, where $Q \geq 0$, so

$$1 \leq Q(0) \leq \left(2^G |\Lambda'| \max_{j \in \Lambda'} \prod_g B_g(j_g)\right) \mathbb{E}Q(K).$$

(If $z_g = 0$ then $j_g = 0$ on Λ' and $B_g(0) = 1$.)

Step 5 (bounds). Apply (2) with $s = s_g$ and $\mu' = q_g z_g$, using $\sum_g j_g s_g \leq \lambda$ and $|\Lambda| \leq (1 + \lambda/s_*)^G$: $\log(1/\mathbb{E}_u f_w) \leq \Phi(w)$, where $\Phi(w) = 19\alpha\lambda + \sum_g [C_4 q_g z_g e^{-2\alpha s_g} +$

$\frac{1}{2} \log(16q_g z_g + 16) + 74] + G \log 2 + G \log(1 + \lambda/s_*)$. By Jensen, $\mathbb{E}f = \mathbb{E}_w \mathbb{E}_u f_w \geq e^{-\mathbb{E}_w \Phi(w)}$. Finally $\mathbb{E}_w q_g z_g = \mu_g := \sum_{i \in B_g} p_i$, concavity of $\log$, $e^{-2\alpha s_g} \leq e^{-\alpha s_i}$ on B_g , and $\sum_g \log(16\mu_g + 16) \leq G \log(16\mu + 16)$. $\square$

4.3. The CRT form.

Theorem 4.5 (sieve limit for prime-slice systems; [ET, Thm 2.5]; [PROVED]). *Take a prime-slice system. Assume that for every $c \in R$, $|F_\ell(c)| \leq \ell/4$ for every $\ell \in \mathcal{P}$ with $\ell \leq e^\lambda$, and $|F_\ell(c)| < \ell$ for every $\ell \in \mathcal{P}$. Let $\bar{\mu} = \sum_{\ell \in \mathcal{P}, \ell \leq e^\lambda} \bar{p}_\ell$ be the truncated mass, $s_* = \log \min \mathcal{P}$, and $G = \lfloor \log_2(\lambda/s_*) \rfloor + 1$. Then every majorant ν of $\mathcal{A}$ of level $\lambda \geq s_*$ satisfies, for every $\alpha > 0$,*

$$\begin{aligned} \log \frac{1}{\mathbb{E}\nu} &\leq \log \frac{Q_0}{|R|} + 19\alpha\lambda + C_4 \sum_{\ell \in \mathcal{P}, \ell \leq e^\lambda} \bar{p}_\ell \ell^{-\alpha} \\ &\quad + G(75 + \log(2 + \lambda/s_*)) + \frac{G}{2} \log(16\bar{\mu} + 16). \end{aligned} \quad (4)$$

If $\lambda < s_*$, ν depends only on the small residue and the bound holds without the λ -terms.

Proof. Use the uniform measure on $\mathbb{Z}/Q_{\text{tot}}$ for a common period. By CRT the coordinates $n \bmod Q_0^*$ (the Q_0 -primary part of Q_{tot}), $n \bmod \ell^{e_\ell}$ ($\ell \in \mathcal{P}$) and the rest are independent and uniform. Since $\nu \geq 0$, $\mathbb{E}\nu \geq (|R|/Q_0) \text{avg}_{c \in R} \mathbb{E}[\nu \mid n \bmod Q_0 = c]$. Fix $c \in R$. The hit indicators $x_\ell = \mathbf{1}[n \bmod \ell \in F_\ell(c)]$ are independent Bern($p_\ell(c)$), independent of the finer digits. Each term of ν , conditioned on (c, x) , depends only on x_{T_i} with $T_i = \{\ell \in \mathcal{P} : \ell \mid d_i\}$, so $\nu_c = \mathbb{E}[\nu \mid c, x]$ is λ -level with weights $s_\ell = \log \ell$, and $\nu_c \geq 0$. The event $\{c\} \times \{x = 0\}$ has positive probability and lies in $\mathcal{A}$, so $\nu_c(0) \geq 1$. Apply Proposition 4.4 in each fibre, then Jensen over $c \in R$; the right side of (3) is linear in the p_ℓ except for the concave logarithmic term. $\square$

Remark 4.6 (size of the error terms). The last two terms of (4) are $O(\log^2 \lambda)$ when $s_* \gg 1$ and $\log \bar{\mu} \ll \log \lambda$. In general $\bar{\mu}$ may be as large as $\pi(e^\lambda)/4$, and then they are $\asymp \lambda \log \lambda$. In every application below $\bar{\mu} \ll \lambda^3$.

Remark 4.7 (the R -term). The term $\log(Q_0/|R|)$ is a saving that the theorem leaves uncontrolled: $\nu = \mathbf{1}_R$ is a majorant with no slice cost and saving exactly $\log(Q_0/|R|)$. For a selector $R = \{(c, P) = 1\}$ the term is $\log(P/\varphi(P)) \leq \log \log \log P + O(1)$. If R itself encodes forced classes with moduli dividing Q_0 , the term is the void probability of a non-slice subsystem, about which Theorem 4.5 says nothing.

Define the *Rankin functional*

$$\Psi(\lambda) = \inf_{\alpha > 0} \left[\alpha\lambda + \sum_{\ell} \bar{p}_\ell \ell^{-\alpha} \right]. \quad (5)$$

Theorem 4.5 says that no majorant of level λ saves more than $C\Psi(\lambda) + O(\log^2 \lambda)$ beyond the R -term, in the situation of Remark 4.6. No converse is claimed in general; attainability is discussed only for the two campaign families (Section 8).

Remark 4.8 (Montgomery's large sieve; [ET, Remark 2.6]; [PROVED]). Montgomery's large sieve [M] applied in a fibre c gives a bound $Y/S_c(Q)$ with $Q^2 \leq Y$, where $S_c(Q) = \sum_{q \leq Q} \mu^2(q) \prod_{\ell \mid q} g_\ell$ and $g_\ell = |F_\ell(c)|/(\ell - |F_\ell(c)|)$. Rankin's trick gives $S_c(Q) \leq Q^\alpha \prod_{\ell} (1 + g_\ell \ell^{-\alpha}) \leq \exp\{\alpha \log Q + 2 \sum_{\ell} p_\ell(c) \ell^{-\alpha}\}$ when $p_\ell(c) \leq 1/2$. Summed over fibres, Jensen gives $\sum_c Y e^{-\Phi_c} \geq |R| Y e^{-\text{avg } \Phi_c}$. So the large-sieve

bound over a slice system obeys the same functional, with level $\lambda = \log Q$, directly and without duality. No sequential or shared-prime large-sieve statement is made.

4.4. Sequential windows: shared large primes. Theorem 4.5 needs the slice primes to be disjoint from the small modulus. The following extension allows a large prime to be the slice prime of one condition and a cofactor prime of another.

Setting [ET, §2.6]. Take $Q_0, R \subseteq \mathbb{Z}/Q_0$, and a set $\mathcal{P}$ of primes not dividing Q_0 , split into ordered windows $\mathcal{P} = W_1 \sqcup \cdots \sqcup W_J$. A *condition* C is an event

$$E_C = \{n \equiv a_C \pmod{q_C}, n \equiv b_{C,\ell} \pmod{\ell^{e_{C,\ell}}} \text{ for } \ell \in S(C)\}$$

with $q_C \mid Q_0$, $\emptyset \neq S(C) \subseteq \mathcal{P}$ and $e_{C,\ell} \geq 1$. Assume

- (U) the last window meeting $S(C)$ meets it in exactly one prime $\ell(C)$, and $e_{C,\ell(C)} = 1$.

Fix $E_\ell \geq \max_C e_{C,\ell}$. The *history* before window j is $H_{<j} = (n \bmod Q_0, (n \bmod \ell^{E_\ell})_{\ell \in W_{<j}})$. For $\ell \in W_j$ and a history h let $F_\ell(h)$ be the set of $b_{C,\ell}$ over the conditions C with $\ell(C) = \ell$ whose other requirements are met by h , and $p_\ell(h) = |F_\ell(h)|/\ell$. The avoider set is $\mathcal{A} = \{n : n \bmod Q_0 \in R, n \notin E_C \ \forall C\}$. Let Q_{seq} be the law of the history built window by window: $c = n \bmod Q_0$ uniform on R ; given $H_{<j}$, the residues $n \bmod \ell$ ($\ell \in W_j$) independent and uniform on $\mathbb{Z}/\ell \setminus F_\ell(H_{<j})$, with uniform independent higher digits. Majorants of level λ charge all primes of $\mathcal{P}$.

Theorem 4.9 (sequential sieve limit; [ET, Thm 2.7]; [PROVED]). *Assume that on the support of Q_{seq} , $p_\ell(h) \leq 1/4$ whenever $\log \ell \leq \lambda$, and $p_\ell(h) < 1$ always. Then for all $\alpha_1, \dots, \alpha_J > 0$, every majorant ν of level λ satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \frac{Q_0}{|R|} + \sum_j \mathbb{E}_{Q_{\text{seq}}} [\Phi_j(H_{<j})],$$

where $\Phi_j(h)$ is the right side of (3) for the coordinates $\ell \in W_j$ with $\log \ell \leq \lambda$, probabilities $p_\ell(h)$ and $\alpha = \alpha_j$. A window with $\log \min W_j > \lambda$ contributes 0.

Proof. Let $g_j(h) = \mathbb{E}[\nu \mid H_{<j} = h]$ under the uniform measure, and let $\mathcal{A}_{<j}$ be the histories in R satisfying no condition whose last window is before j . We show by downward induction that for $h \in \mathcal{A}_{<j}$, $-\log g_j(h) \leq \mathbb{E}_{Q_{\text{seq}}} [\sum_{i \geq j} \Phi_i(H_{<i}) \mid H_{<j} = h]$. For $j = J+1$ the history decides membership in $\mathcal{A}$, and $\nu \geq 1$ on $\mathcal{A}$. For the step, fix $h \in \mathcal{A}_{<j}$. Given h , the residues $n \bmod \ell^{E_\ell}$, $\ell \in W_j$, are independent and uniform, so the indicators $x_\ell = \mathbf{1}[n \bmod \ell \in F_\ell(h)]$ are independent $\text{Bern}(p_\ell(h))$. Put $f(x) = \mathbb{E}[g_{j+1}(H_{<j+1}) \mid h, x]$. Then $\mathbb{E}f = g_j(h)$, $f \geq 0$, and f is λ -level on W_j . The event $x = 0$ is exactly that no condition with last window j holds; given it, $H_{<j+1} \in \mathcal{A}_{<j+1}$ with the Q_{seq} transition law. The induction hypothesis and Jensen give $f(0) > 0$ and the bound for $f(0)$; Proposition 4.4 for $f/f(0)$ gives $\mathbb{E}f \geq f(0)e^{-\Phi_j(h)}$. Finally $\mathbb{E}\nu \geq (|R|/Q_0) \text{avg}_{c \in R} g_1(c)$ and Jensen over c . $\square$

Lemma 4.10 (profile under Q_{seq} ; [ET, Lemma 2.8]; [PROVED]). *For $\ell \in W_j$,*

$$\mathbb{E}_{Q_{\text{seq}}} p_\ell(H_{<j}) \leq \sum_{C: \ell(C)=\ell} \frac{1}{\ell} \mathbb{P}(c \equiv a_C \pmod{q_C} \mid c \in R) \prod_{\ell' \in S(C) \setminus \{\ell\}} \frac{1}{\ell'^{e_{C,\ell'}} (1 - p_{\ell'}^*)},$$

where $p_{\ell'}^*$ is the supremum of $p_{\ell'}(h)$ over reachable histories.

Proof. Chain rule over windows: given the past, $n \bmod \ell'$ is uniform on at least $\ell'(1 - p_{\ell'}^*)$ residues, and the higher digits are uniform. $\square$

5. SUPPLY PROFILES

We use Shiu's theorem [S, Theorem 1] in the form stated in [TQ, §3]: if $F \geq 0$ is multiplicative, $F(p^a) \leq A_1^a$, and $F(n) \ll_\epsilon n^\epsilon$, then for fixed $0 < \alpha_s, \beta_s < 1/2$,

$$\sum_{\substack{V-Y < n \leq V \\ n \equiv a \pmod{q}}} F(n) \ll \frac{Y}{\varphi(q) \log V} \exp \left\{ \sum_{p \leq V, p \nmid q} \frac{F(p)}{p} \right\}$$

uniformly for $q \geq 2$, $(a, q) = 1$, $q < Y^{1-\alpha_s}$, $V^{\beta_s} < Y \leq V$.

Lemma 5.1 (cubic Case-B profile, $\mathcal{R}(M)$; [ET, Lemma 3.1]; [PROVED]). *Put $S_B(x) = \sum_{M \leq x, M \equiv 3 \pmod{4}} \tau(A^2)M/\varphi(M)$ with $A = (M+1)/4$. Then $S_B(x) \ll x(\log 2x)^2$. Consequently, for $0 < \beta \leq 1$, $\sum_{M \equiv 3 \pmod{4}} \tau(A^2)(M/\varphi(M))M^{-1-\beta} \leq C_6\beta^{-3}$.*

Proof. Write $M/\varphi(M) = \sum_{d|M} \mu^2(d)/\varphi(d)$ and swap sums; the inner sum runs over $A \leq (x+1)/4$ with $4A \equiv 1 \pmod{d}$. For $2 \leq d \leq x^{1/2}$, Shiu with $F(A) = \tau(A^2)$ ($F(p) = 3$) in the reduced class $4^{-1} \pmod{d}$ gives $\ll (x/\varphi(d))(\log x)^2$; for $d = 1$ the elementary bound $\sum_{A \leq x} \tau(A^2) \ll x \log^2 x$ suffices; and $\sum_d \varphi(d)^{-2} < \infty$. For $d > x^{1/2}$ bound trivially by $(x/d+1)x^{1/10}$, for a total $\ll x^{0.61}$. The β -bound follows by partial summation, $\int (\log 2x)^2 x^{-1-\beta} dx \ll \beta^{-3}$. $\square$

Numerically $S_B(x)/(x \log^2 x) = 0.1067, 0.0990, 0.0963$ at $x = 10^2, 10^4, 10^6$ [EVIDENCE].

Lemma 5.2 ($((a, D)$ profile; [ET, Lemma 3.2]; [PROVED]). *With $G = 4a g(D)$, $\sum_{a,D} (G/\varphi(G))G^{-1-\beta} \ll \beta^{-3}$ for $0 < \beta \leq 1$.*

Proof. Using $\#\{D : g(D) = g\} \leq 2^{\omega(g)}$, the sum is at most

$$\ll \left(\sum_a a^{-1-\beta} \frac{a}{\varphi(a)} \right) \left(\sum_g 2^{\omega(g)} \frac{g}{\varphi(g)} g^{-1-\beta} \right) \ll \beta^{-1} \beta^{-2}.$$

$\square$

Lemma 5.3 (weighted Case-A divisor sum; [ET, Lemma 3.7]; [PROVED MOD] Elsholtz–Tao Prop. 1.4). *For $x \geq 2$,*

$$\sum_{r,h \geq 1, rh \leq x} \tau(4rh^2 + 1) \frac{rh}{\varphi(rh)} \ll x(\log x)^2. \quad (6)$$

The same holds with the extra weight $\gamma(rh)$ of Corollary 6.2 below. Consequently the Case-A classes $-m^{-1} \pmod{4g(d)}$, with the selector weight, have profile $\ll \beta^{-3}$.

Proof. The input is [EIT, Prop. 1.4]: for $A, B > 1$ and every positive integer $k \ll (AB)^{O(1)}$, $\sum_{a \leq A} \sum_{b \leq B} \tau(kab^2 + 1) \ll AB \log(A+B) \log(1+k)$. Use $n/\varphi(n) \leq (\pi^2/6) \sum_{d|n} 1/d$ and write $r = sr'$, $h = th'$, so that $4rh^2 = (4st^2)r'h'^2$. The left side of (6) is at most $C \sum_{s,t} (st)^{-1} \sum_{r'h' \leq Y} \tau(4st^2r'h'^2 + 1)$ with $Y = x/(st)$. If $st \leq \sqrt{x}$, cover $r'h' \leq Y$ by $O(\log Y)$ rectangles with $AB \asymp Y \geq \sqrt{x}/2$, $A, B \geq 2$; then $k = 4st^2 \ll (AB)^3$, and Prop. 1.4 gives $\ll Y(\log x)^2 \log(1 + 4st^2)$; finally $\sum_{s,t} \log(1 + 4st^2)/(st)^2 < \infty$. If $st > \sqrt{x}$ then $Y < \sqrt{x}$ and $\tau \ll x^\epsilon$ give $\ll x^{1/2+2\epsilon}$. For the γ -weight replace $1/(st)$ by $\eta(s)\eta(t)$, where $\gamma(n)n/\varphi(n) = \sum_{d|n} \eta(d)$ and $\eta(p) = O(p^{-1} + p^{-\delta})$; the needed sums converge because $\eta(p)/p \ll p^{-1-\delta}$ and

$\sum_n \eta(n) n^{-1+\delta/2} < \infty$. The profile statement follows by partial summation, since the Case-A class mass up to modulus G is at most $\sum_{4rh \leq G} \tau(4rh^2 + 1)/(4rh)$. $\square$

The review [ETr, item 4] checked Prop. 1.4 and its k -range against the source; it was not re-proved.

6. THE THREE-QUARTER CAP

6.1. Prime-slice families.

Corollary 6.1 ([ET, Cor. 3.4]; [PROVED]). *Fix $0 < C < 1$ and $A \geq 1$. Take a prime-slice system in which every condition is a slice condition (no pure small-modulus conditions), and*

- *every condition is a forced class of $\mathcal{R}(M)$ (any splitting $M = q_0\ell$), an (a, D) -class, or a Case-A class;*
- *each condition modulus $q_0\ell$ satisfies $q_0 \leq \ell^C$;*
- *$R = \{c : (c, P) = 1\}$ for some $P \mid Q_0$;*
- *$\ell \geq \ell_0(C)$ for every slice prime.*

Then every majorant of level $\lambda \leq A \log N$ satisfies

$$\log \frac{1}{\mathbb{E}_\nu} \leq C_7(A, C)(\log N)^{3/4} + \log \frac{P}{\varphi(P)},$$

and so does the large-sieve bound of Remark 4.8, summed over fibres.

Proof. Since $q_0 \leq \ell^C$ and the number of family classes modulo M is $M^{o(1)}$, $|F_\ell(c)| \leq \ell^{C+o(1)} \leq \ell/4$. For a selector, $\mathbb{P}(c \equiv b(q_0) \mid R) \leq 1/\varphi(q_0)$, so $\bar{p}_\ell \leq \sum_{M=q_0\ell} |\mathcal{C}(M)|(M/\varphi(M))/M$, where $|\mathcal{C}(M)|$ is the number of family classes modulo M . Since $\ell \geq M^{1/(1+C)}$, $\ell^{-\alpha} \leq M^{-\alpha/2}$. Lemmas 5.1, 5.2 and 5.3 give $\sum_\ell \bar{p}_\ell \ell^{-\alpha} \ll \alpha^{-3}$. Take $\alpha = \lambda^{-1/4}$ in (4); here $G = O(\log \lambda)$ and $\bar{\mu} \ll \lambda^3$. $\square$

6.2. Dominant-prime families.

Corollary 6.2 ([ET, Cor. 3.6]; [PROVED]). *Fix $0 < C < 1$ and $A \geq 1$, and put $\delta = (1-C)/2$. Consider any family of Case-B forced classes (classes of $\mathcal{R}(M)$, or (a, D) -classes modulo $G = 4a g(D)$), and Case-A classes (modulo $G = 4rh$), such that every modulus M has a prime factor $\ell(M) \geq M^{1/(1+C)}$. The other prime factors are unrestricted: they may be large, occur to higher powers, or be the dominant prime of other moduli. Let $w_0 = w_0(C)$ be large, $Q_0 = \text{lcm}(P_{w_0}, w_0\text{-smooth parts of all moduli})$ and $R = \{c : (c, P_{w_0}) = 1, c \text{ satisfies no condition with all primes } \leq w_0\}$. Then every majorant of level $\lambda \leq A \log N$, charging all primes $> w_0$, satisfies*

$$\log \frac{1}{\mathbb{E}_\nu} \leq C_8(A, C)(\log N)^{3/4} + \log \frac{Q_0}{|R|}, \quad \log \frac{Q_0}{|R|} = O_C(1).$$

Proof. The R-term. A w_0 -smooth modulus is at most w_0^{1+C} , so there are finitely many. Let $L' = \text{lcm}(P_{w_0}, w_0\text{-smooth moduli})$. By Lemma 2.5, $c \equiv 1 \pmod{L'}$ lies in R , so $|R|/Q_0 \geq 1/L' = e^{-O_C(1)}$.

Windows. With $s_0 = \log w_0$ take $W_j = (e^{s_j}, e^{s_j/C}]$, $s_j = s_0 C^{-j+1}$, and attach each condition to $\ell(M)$. Since $\ell(M) > \sqrt{M}$ it occurs to exponent one, and every other prime $\ell' \mid M$ has $\ell' \leq M/\ell \leq \ell^C$, so it lies in an earlier window or divides Q_0 . Thus (U) holds.

Hypotheses of Theorem 4.9. The number of conditions with dominant prime ℓ is at most $\sum_{q \leq \ell^C} \ell^{o(1)} = \ell^{C+o(1)}$, so $p_\ell^* \leq \ell^{-\delta} \leq 1/4$ for $\ell > w_0$.

Profile. By Lemma 2.5, $\mathbb{P}(c \equiv a(q) \mid R) \leq L'/\varphi(q)$. By Lemma 4.10 each condition contributes at most $L'(M/\varphi(M))\gamma(M)/M$ with $\gamma(M) = \prod_{\ell' \mid M, \ell' > w_0} (1 - \ell'^{-\delta})^{-1}$. Write $(M/\varphi(M))\gamma(M) = \sum_{d \mid M} h(d)$, h multiplicative on squarefree d with $h(p) \leq 1/(p-1) + 3p^{-\delta}$. The proof of Lemma 5.1 goes through: in the Shiu range it uses $\sum_d h(d)/\varphi(d) < \infty$; for $d > x^{1/2}$, $\sum_d h(d)d^{-1+\delta/2} < \infty$ gives a tail $\ll x^{1-\delta/4+\epsilon} = o(x)$. For (a, D) -classes use $(G/\varphi(G))\gamma(G) \leq 2[(a/\varphi(a))\gamma(a)][(g/\varphi(g))\gamma(g)]$, and for Case A use the γ -weighted form of Lemma 5.3. So the profile is $\ll_C \beta^{-3}$.

Summing windows. For $\ell \in W_j$, $M \leq \ell^{1+C} \leq e^{2s_j/C}$, so the window mass is $\ll s_j^3$ and its profile is $\ll e^{-\alpha s_j} s_j^3$. Optimising α_j per window, $\mathbb{E}\Phi_j \ll \min\{s_j^3, (\lambda/s_j)(1 + \log^+(s_j^4/\lambda))\} + O(\log^2 \lambda)$. Only $J \ll_C \log \lambda$ windows meet $\ell \leq e^\lambda$. Below the crossover $s \approx \lambda^{1/4}$ the cubic branch is a geometric sum; above it, with $s = \lambda^{1/4} C^{-k}$, the window contributes $O_C(\lambda^{3/4} C^k(1+k))$; the remainders total $O_C(\log^3 \lambda)$. Hence $\sum_j \mathbb{E}\Phi_j \ll_C \lambda^{3/4}$. $\square$

The constants deteriorate as $C \rightarrow 1$. For unweighted integers, Dickman's function gives logarithmic density $\log(1+C)$ to the moduli with $P(M) \geq M^{1/(1+C)}$; the share of the *weighted* cubic supply is not proved [CONJECTURE].

6.3. The main theorem.

Theorem 6.3 (the 3/4 cap; [PROVED] mod Elsholtz–Tao Prop. 1.4 for Case-A classes; unconditional for Case-B families). *Fix $0 < C < 1$ and $A \geq 1$. Let $\mathfrak{F}$ be a family of forced classes such that either*

- (i) *every modulus has a dominant prime $P(M) \geq M^{1/(1+C)}$ (Case-B and Case-A classes; Corollary 6.2); or*
- (ii) *$\mathfrak{F}$ consists of $\mathcal{R}(M)$ -classes with (η, B) -gapped moduli, for fixed B and $0 < \eta < 1$, together with any $\mathcal{R}(M)$ -classes of $W_0(B)$ -smooth modulus (Theorem 7.6, Section 7).*

Then every CRT-majorant architecture (Definition 3.3) for $\mathfrak{F}$, with the admissible set of Corollary 6.2 in case (i) and the quadratic-residue base of Section 7 in case (ii), saves at most

$$\log \frac{1}{\mathbb{E}\nu} \ll (\log N)^{3/4},$$

with implied constant depending on A, C in case (i) and on A, B, η in case (ii). In particular no such architecture gives $E(N) \leq N \exp\{-(\log N)^{3/4} \omega(N)\}$ with $\omega(N) \rightarrow \infty$, e.g. $\omega = (\log \log N)^\epsilon$.

Proof. Corollary 3.5, applied with Corollary 6.2 in case (i) and Theorem 7.6 in case (ii). In both the R -term is bounded. $\square$

The statement for the prime-slice families of Corollary 6.1 is the same, with the selector term $\log(P/\varphi(P))$ added; for the large sieve over slice systems the level hypothesis $\lambda \leq A \log N$ is used directly (Remark 4.8).

7. GAPPED MODULI, UNCONDITIONALLY

Corollary 6.2 needs $P(M) \geq M^{1/(1+C)}$ with $C < 1$. This section treats a class of *balanced* moduli ($P(M) \leq M^{1/2}$) as well, for the classes $\mathcal{R}(M)$. The results are from [TW, §§1–2, 4], building on the reductions of [EB, §2]; the review [TW_r] (two rounds) found them sound.

7.1. Gapped and twin moduli. Write $P_2(M) = P(M/P(M))$, with $P_2(1) = 1$. Fix $w_0 \geq 3$, $0 < \eta < 1$ and $B \geq 0$. Following [EB, §2], M is η -gapped if $P(M) > w_0$ and $P(M) \geq P_2(M)^{1+\eta}$, and η -twin otherwise; it is (η, B) -gapped if moreover $M \leq P(M)^{1+B}$. Put $s_0 = \log w_0$, $s_j = s_0(1 + \eta)^{j-1}$ and use the windows $W_j = \{\ell \text{ prime} : s_j < \log \ell \leq s_{j+1}\}$.

Lemma 7.1 (gapped implies (U); [EB, Lemma 2.1]; [PROVED]). *If M is η -gapped and $\ell = P(M)$, then $\ell \parallel M$, and every prime $\ell' \mid M/\ell$ with $\ell' > w_0$ lies in a window strictly before that of ℓ . So a family of η -gapped classes satisfies (U) with $\ell(C) = P(M_C)$.*

Proof. If $\ell^2 \mid M$ then $P_2(M) = \ell$, contradicting $\ell \geq \ell^{1+\eta}$. If $\ell' \in W_j$ divides M/ℓ , then $\log \ell \geq (1 + \eta) \log \ell' > (1 + \eta)s_j = s_{j+1}$. $\square$

Dominant moduli with $P(M) > w_0$ are η -gapped for every $\eta \leq \min(\eta_0, 1/C - 1)$. Balanced moduli $M = k\ell_1\ell_2$ with $P(k) < \ell_1$, $\ell_2 \geq \ell_1^{1+\eta}$ and $k\ell_1 \geq \ell_2$ are η -gapped. The same-scale pairs of Lemma 9.1 below are η -twin. For a gapped balanced modulus the cofactor $M/P(M)$ exceeds $P(M)$, so the counting bound $p_\ell \leq \ell^{C-1+o(1)}$ of Corollary 6.2 is lost; the hit probabilities can be large for some histories (*heavy coordinates*).

7.2. The quadratic-residue base. Fix $W \geq 3$, let Q_0 be a common multiple of the W -smooth parts of all moduli with $P_W \mid Q_0$, and put $R_W = \{c \bmod Q_0 : (\frac{c}{p}) = 1 \text{ for every odd prime } p \leq W\}$.

Lemma 7.2 (QR base; [TW, Lemma 1.3]; [PROVED]). *(1) Every $c \in R_W$ avoids every class of $\mathcal{R}(M)$ with W -smooth M . (2) $\log(Q_0/|R_W|) = \sum_{3 \leq p \leq W} \log(2p/(p-1))$. (3) Under the uniform law on R_W , residues at distinct primes are independent and $\mathbb{P}(c \equiv a \pmod{p^e}) \leq \gamma(p)/p^e$ for $p^e \mid Q_0$, with $\gamma(p) = 2p/(p-1)$ for odd p and $\gamma(2) = 1$.*

Proof. (1) is Lemma 2.6. (2) and (3): by CRT R_W is a product over the prime powers $p^e \parallel Q_0$; for odd $p \leq W$ the factor has $p^{e-1}(p-1)/2$ elements out of p^e . $\square$

So Mordell's obstruction, which classically shows that polynomial identities never cover squares, is used constructively: it gives a *product* base measure on avoiders of all small-modulus classes, with polylogarithmic inflation.

7.3. The capped sequential measure. Work in the sequential setting of Section 4.4, with R satisfying (R1) every $c \in R$ avoids every condition with $S(C) = \emptyset$, and (R2) under the uniform law on R the residues at distinct primes are independent and $\mathbb{P}(c \equiv a \pmod{p^e}) \leq \gamma(p)/p^e$. Fix caps $\delta_\ell \in (0, 1/4]$. The *capped measure* Q' draws c uniformly from R ; given $H_{<j} = h$, the residues $n \bmod \ell$ ($\ell \in W_j$) are independent, uniform on $\mathbb{Z}/\ell \setminus F_\ell(h)$ if $p_\ell(h) \leq \delta_\ell$ (*light*), and uniform on $\mathbb{Z}/\ell$ otherwise (*heavy*); higher digits are uniform. The *leak* is $\mathfrak{L} = Q'(\text{final history} \notin \mathcal{A})$. This is a variant of the distortion method of Hough [H] and Balister–Bollobás–Morris–Sahasrabudhe–Tiba [BBMST, def. (5), Thm 3.1], adapted to general level- λ majorants.

Lemma 7.3 (leak and inflation; [TW, Lemmas 2.1, 2.2]; [PROVED]). *(a) $\mathfrak{L} \leq \sum_{\ell \in \mathcal{P}} \mathbb{E}_{Q'}[p_\ell(H_{<j(\ell)}) \mathbf{1}\{p_\ell(H_{<j(\ell)}) > \delta_\ell\}]$. (b) For m a product of prime powers p^e with $e \leq E_p$ ($p \in \mathcal{P}$) or $p^e \mid Q_0$, and every a , $Q'(n \equiv a \pmod{m}) \leq \prod_{p^e \parallel m} \gamma'(p)/p^e$, where $\gamma'(p) = \gamma(p)$ for $p \mid Q_0$ and $\gamma'(p) = (1 - \delta_p)^{-1}$ for $p \in \mathcal{P}$.*

Proof. (a) A final history outside $\mathcal{A}$ satisfies some condition C with $S(C) \neq \emptyset$, by (R1). At the transition into the window of $\ell = \ell(C)$, $b_{C,\ell} \in F_\ell(h)$, and light coordinates never take such values, so ℓ was heavy and the hit has probability $p_\ell(h)$; take a union bound. (b) Chain rule: (R2) at the base; a light coordinate is uniform on at least $p(1 - \delta_p)$ residues, a heavy one on all p . $\square$

Theorem 7.4 (capped sequential sieve limit; [TW, Thm 2.3]; [PROVED]). *If $\mathfrak{L} \leq 1/2$, every majorant ν of level λ of $\mathcal{A}$ satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \frac{Q_0}{|R|} + \log 2 + 2 \sum_j \mathbb{E}_{Q'} [\Phi_j^{\text{light}}(H_{<j})],$$

where $\Phi_j^{\text{light}}(h)$ is the right side of (3) for the coordinates of W_j that are light at h and have $\log \ell \leq \lambda$.

Proof. Let $g_j(h) = \mathbb{E}_U[\nu \mid H_{<j} = h]$ under the uniform measure. By downward induction, for every history h at window j ,

$$g_j(h) \geq \mathbb{E}_{Q'} \left[\mathbf{1}_{\mathcal{A}}(H_{<j+1}) \exp \left(- \sum_{i \geq j} \Phi_i^{\text{light}}(H_{<i}) \right) \mid H_{<j} = h \right].$$

The base case is $g_{J+1} \geq \mathbf{1}_{\mathcal{A}}$. For the step, the hit indicators under U given h are independent $\text{Bern}(p_\ell(h))$; split $x = (x_L, x_H)$ into light and heavy and put $\tilde{f}(x_L) = \mathbb{E}_U[g_{j+1}(H_{<j+1}) \mid h, x_L]$. Then $\mathbb{E}\tilde{f} = g_j(h)$, $\tilde{f} \geq 0$, $\tilde{f}$ is λ -level in x_L (averaging over x_H preserves this), light coordinates have $p_\ell(h) \leq 1/4$, and given $x_L = 0$ the law of $H_{<j+1}$ is the Q' transition. If $\tilde{f}(0) = 0$ the claim is trivial; otherwise Proposition 4.4 for $\tilde{f}/\tilde{f}(0)$ gives $\mathbb{E}\tilde{f} \geq \tilde{f}(0)e^{-\Phi_j^{\text{light}}(h)}$, and the induction hypothesis bounds $\tilde{f}(0)$. Finally, with $S = \sum_j \Phi_j^{\text{light}} \geq 0$, $\mathbb{E}\nu \geq (|R|/Q_0)\mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}}e^{-S}]$, and Jensen on $Q'(\cdot \mid \mathcal{A})$ with $Q'(\mathcal{A}) \geq 1/2$ gives $\mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}}e^{-S}] \geq \frac{1}{2}e^{-2\mathbb{E}_{Q'}S}$. $\square$

Heavy coordinates thus cost nothing in the bound; they enter only through $\mathfrak{L}$, which is controlled by an average, never by a supremum.

7.4. The gapped cap. Take the windows above with $w_0 = W$, the base $R = R_W$, and caps $\delta_\ell = \ell^{-1/2}$. The family $\mathfrak{G}$ consists of $\mathcal{R}(M)$ -classes with M (η, B)-gapped and $P(M) > W$, and any $\mathcal{R}(M)$ -classes with W -smooth M . The latter are avoided by R_W (Lemma 7.2), so (R1) holds; (U) holds by Lemma 7.1. Then $\Gamma(m) := \prod_{p|m} \gamma'(p) \leq 3^{\omega(m)}$.

Lemma 7.5 (second moment; [TW, Lemma 2.4]; [PROVED]). *For every $\epsilon > 0$ there is $C(\epsilon, B)$, independent of W , η and the family, such that $\mathbb{E}_{Q'}[p_\ell(H)^2] \leq C(\epsilon, B)\ell^{-2+\epsilon}$ for every prime $\ell > W$.*

Proof. $|F_\ell(h)|$ is at most the number $N_\ell(n)$ of pairs (q, D) with $q \leq \ell^B$, $D \mid A_{q\ell}^2$ and $n \equiv -4D \pmod{q}$, a function of the full history. Expanding N_ℓ^2 , a compatible pair of pairs contributes $Q'(n \equiv a \pmod{m}) \leq 3^{\omega(m)}/m$ with $m = \text{lcm}(q, q')$, by Lemma 7.3(b). Each q carries $\tau(A^2) \ll_\epsilon \ell^{\epsilon/4}$ divisors D , and there are $\tau(m^2)$ pairs with $\text{lcm} = m$. So $\mathbb{E}N_\ell^2 \ll_\epsilon \ell^{\epsilon/2} \sum_{m \leq \ell^{2B}} 3^{\omega(m)} \tau(m^2)/m \ll_{\epsilon, B} \ell^\epsilon$. Divide by ℓ^2 . $\square$

By Lemma 7.3(a) and Markov, $\mathfrak{L} \leq \sum_{\ell > W} \ell^{1/2} C(1/4, B) \ell^{-7/4} \ll_B W^{-1/4}$, so there is $W_0(B)$ with $\mathfrak{L} \leq 1/2$ for $W \geq W_0(B)$, uniformly in η and the family [TW,

Cor. 2.5]. The same weighted Shiu argument as in Corollary 6.2 (now with $h(p) \leq 2$ for $p \leq W$ and $h(\ell) \leq 2\ell^{-1/2}$ for $\ell > W$) gives, for every window with $s_j \leq \lambda$,

$$\mathbb{E}_{Q'} \sum_{\ell \in W_j} p_\ell(H_{<j}) \leq K((1+B)s_j)^3, \quad K = K(W, B) \quad (7)$$

[TW, Lemma 2.6], using $M \leq P(M)^{1+B} \leq e^{(1+B)s_{j+1}}$ [EB, Lemma 2.4].

Theorem 7.6 (the gapped cap; [TW, Thm 2.7]; [PROVED]). *Fix $B \geq 0$ and take $W = W_0(B)$. For every $0 < \eta < 1$, every family $\mathfrak{G}$ as above, and every majorant ν of level $\lambda \geq \log W$,*

$$\log \frac{1}{\mathbb{E}_\nu} \leq C K^{1/4} (1+B)^{3/4} \eta^{-1} \lambda^{3/4} + O_B(\eta^{-1} \log^2 \lambda) + O_B(1),$$

with C absolute and $K = K(W_0(B), B)$.

Proof. Theorem 7.4 applies since $\mathfrak{L} \leq 1/2$; the R -term is $O_B(1)$ by Lemma 7.2(2). Within a window the costs lie in $(s_j, (1+\eta)s_j]$ and $1+\eta < 2$, so after Step 0 of Proposition 4.4 there is one nonempty band, and $G = 1$ with s_* the least cost present [EB, Lemma 2.5a]. The light profile is at most the full profile, so (7) and concavity bound the logarithmic terms by $O(\log \lambda + \log K + \log(1+B))$ per window, and there are at most $1 + 2 \log \lambda / \eta$ windows with $s_j \leq \lambda$. The main term of window j is at most $\inf_\alpha [19\alpha\lambda + X_j e^{-\alpha s_j}] \leq \min\{X_j, (19\lambda/s_j)(1 + \log^+(X_j s_j / 19\lambda))\}$ with $X_j = C_4 K (1+B)^3 s_j^3$. Put $s^* = (19\lambda / (C_4 K (1+B)^3))^{1/4}$. Windows with $s_j \leq s^*$ contribute at most $2X(s^*)/\eta = 38\lambda/(\eta s^*)$; the k -th window above s^* contributes at most $(19\lambda/s^*)(1+\eta)^{-k}(1+4(k+1)\log(1+\eta))$, summing to at most $(19\lambda/s^*) \cdot 18/\eta$ [EB, proof of Thm 2.5]. The total is $\leq 380\lambda/(\eta s^*)$. Multiply by 2. $\square$

For dominant moduli this recovers Corollary 6.2 for $\mathcal{R}(M)$ -classes, with a different R -term. It needs no hypothesis on heavy coordinates: the extremal statement (E_δ) and hypothesis H_{light} of [EB, §§2.5, 4.4] are not used.

Scope of Theorem 7.6. It covers only $\mathcal{R}(M)$ -classes: Lemma 2.6 is proved for $\mathcal{R}(M)$ only, and the analysis of [EB] is restricted to $\mathcal{R}(M)$. It needs B fixed: there is no summation over B . It does not cover η -twin moduli, for which (U) fails.

Theorem 7.7 (window-resolved families; [TW, Thm 4.4]; [PROVED]). *Fix B , $W = W_0(B)$ and $\lambda \geq (2 \log W)^4$. Let $\mathfrak{G}$ be any family of $\mathcal{R}(M)$ -classes with $M \leq P(M)^{1+B}$, plus any W -smooth classes, such that every modulus with top prime in $(e^{\lambda^{1/4}}, e^{\lambda/2}]$ is window-resolved: the last η -window meeting its primes $> W$ contains exactly one of them, to exponent one. Then every majorant of level λ satisfies $\log(1/\mathbb{E}_\nu) \ll_B \eta^{-1} \lambda^{3/4}$.*

The proof [TW, §4] uses an abstract form of Theorem 7.4 with blocks: singletons below $e^{\lambda^{1/4}}$, where even twin and prime-power moduli cost $\ll \lambda^{3/4}$ in total [TW, Prop. 4.1]; the η -windows in $(e^{\lambda^{1/4}}, e^{\lambda/2}]$; one block above $e^{\lambda/2}$, where each term of ν sees at most one prime and a linear-window inequality gives cost $O(e^{-\lambda/4})$ [TW, Lemma 4.2, Cor. 4.3]; and a second-moment bound with prime-power top primes [TW, Lemma 4.0] for the leak. So all twin moduli with top prime outside $(e^{\lambda^{1/4}}, e^{\lambda/2}]$ are already covered. The *residual* is the set of unresolved moduli with top prime in that range.

8. THE CAMPAIGN SIEVES LIE IN THE CLASS

8.1. The three-quarter note.

Lemma 8.1 ([ET, Lemma 4.1]; [PROVED]). *The atom family $\mathcal{A}_X$ of [TQ, §2] is a prime-slice system with $Q_0 = \text{lcm}(L_K, P_y)$, $\mathcal{P}$ the primes in $(X^{1/2}, X]$, the selector $R = \{(c, P_y) = 1\}$, $|F_\ell(c)| \leq \ell^{1/3}$, and $\sum_\ell p_\ell(c) \ell^{-\alpha} \leq C_u t^3 X^{-\alpha/2}$ for all c (the upper half of [TQ, Cor. 4.3], “uniform fibre masses”), where $t = \log X$. Its majorant $\nu_X = S_y Q_r(H_X)$ is ≥ 0 on $\mathbb{Z}$, equals 1 on $\mathcal{A}$, and has level $\lambda = rt$. Hence every majorant on this family has*

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \log y + \min \left\{ Ct^3, C \frac{\lambda}{t} \left(1 + \log^+ \frac{Ct^4}{\lambda} \right) \right\} + O(\log^2 \lambda).$$

Proof. The structure is [TQ, Lemma 2.2] (deduplication and conditional independence): conditional on $c \bmod L_K$ the coordinates $n \bmod \ell$ are independent and uniform, and an atom (k, ℓ, u, v) forbids $-uv^{-1} \bmod \ell$ in the fibres with $u + cv \equiv 0 \pmod{k}$; at most $z_j^2 \leq \ell^{1/3}$ atoms live at each ℓ . The majorant is a product of the selector S_y and $Q_r(H) = \binom{H-1}{r}$, both nonnegative, and $Q_r(0) = 1$. A term of $Q_r(H_X)$ involves at most r atoms, hence at most r slice primes $\leq X$, so its level is at most rt . The bound is Theorem 4.5 with α optimised and the selector term $\log(P_y/\varphi(P_y)) = \log \log y + O(1)$. $\square$

With $\lambda \leq A \log N$ the maximum over t is $\asymp (\log N)^{3/4}$, attained only for $t \asymp (\log N)^{1/4}$. What forces $\lambda \lesssim \log N$ is not the note’s modulus bound, which it calls “not necessary”, but its coefficient budget $T_{\text{abs}} \leq N^{1/2}$ [TQ, proof of Thm 8.2]: the slice primes are $\leq X$, so $\Lambda_0 = t \leq \log N$, and Lemma 3.4 converts any majorant on this family with coefficient sum $< N$ into one of level $\leq t + \log N + s$ with at most twice the mean. So the note satisfies (A1)–(A3), and *its exponent 3/4 is sharp for its own architecture* [PROVED]. This was the review’s one substantive correction [ETr, item 3, SC1].

Lemma 8.2 (Bonferroni depth; [ET, Lemma 4.2]; [PROVED]). *For even r and any integer $H \geq 0$, $\mathbb{E}Q_r(H) \geq \mathbb{P}(H \geq r+1)$. If H is a sum of independent Bernoulli variables with mean μ and $r+1 \leq \mu - 2\sqrt{\mu}$, then $\mathbb{E}Q_r(H) \geq 3/4$.*

Proof. $Q_r(h) = \binom{h-1}{r} \geq 1$ for $h \geq r+1$, and Chebyshev with $\text{Var } H \leq \mu$. $\square$

So the depth $r \gg t^3$ in [TQ] is forced. The binding constraint in the 3/4 proof is the pair (cubic supply $\mu_c \leq C_u t^3$, budget $\log T_{\text{abs}} \leq \log N$). Bonferroni depth, the Bombieri–Vinogradov level, the selector and the pruning cannot improve the exponent: the cap of Theorem 4.5 holds for every majorant, and Lemmas 5.1–5.3 bound the *entire* forced-class supply independently of any level of distribution, so Elliott–Halberstam cannot raise the profile either [ET, Lemma 4.3 and remark] [PROVED]. That Selberg or other majorants *attain* the same order up to constants is only [EVIDENCE]: in an exchangeable Poisson model the numerically computed linear-programming optimum agrees with the Selberg value to within 0.01 in $-\log$ in the tested cases [ET, §2.5].

8.2. The two-thirds loglog note. The note [LL] is not a majorant sieve but Montgomery’s large sieve over progressions modulo L_K , with multipliers $\mathcal{K} = \{k \leq \delta \log N\}$. It is covered through Remark 4.8.

Corollary 8.3 (restricted multiplier sets; [ET, Cor. 3.5]; [PROVED] given the notes' Brun–Titchmarsh upper bounds). *Suppose a family at scale $X = e^t$ uses slice primes in $(X^{1/2}, X]$ and multipliers $k \in \mathcal{K}$, with $\sum_{\ell} p_{\ell}(c) \leq A_0 t^2 h(\mathcal{K})$ for all c , where $h(\mathcal{K}) = \sum_{k \in \mathcal{K}} \varphi(k)/k^2$. Then*

$$\text{saving} \leq \min \left\{ Ct^2 h, \frac{38\lambda}{t} \left(1 + \log^+ \frac{Ct^3 h}{\lambda} \right) \right\} + O(\log^2 \lambda),$$

and the supremum over t is $\asymp \lambda^{2/3} h^{1/3}$. For $\mathcal{K} = \{k \leq \delta \log N\}$, $h \asymp \log \log N$, giving $(\log N)^{2/3} (\log \log N)^{1/3}$; the same holds for any $\mathcal{K}$ with $\text{lcm}(\mathcal{K}) \leq N$, since then $h(\mathcal{K}) \leq \prod_{p|\text{lcm}} (1 + 1/p) \ll \log \log N$.

Proof. Insert $\sum_{\ell} \bar{p}_{\ell} \ell^{-\alpha} \leq A_0 t^2 h e^{-\alpha t/2}$ into (4). With $t_0 = (\lambda/h)^{1/3}$, for $t \leq t_0$ the first entry is $\leq C \lambda^{2/3} h^{1/3}$; for $t = ut_0 \geq t_0$ the second is $\leq C \lambda^{2/3} h^{1/3} (1 + 3 \log u)/u$. $\square$

The note's large-sieve level is $Q^2 \leq N/L_{\mathcal{K}}$, so $\lambda \leq \log N$, and its admissible set is the reduced residues modulo $L_{\mathcal{K}}$, with R -term $\log(L_{\mathcal{K}}/\varphi(L_{\mathcal{K}})) \ll \log \log \log N$. By Remark 4.8 and Corollary 8.3 its bound cannot exceed $\exp\{-C(\log N)^{2/3} (\log \log N)^{1/3}\}$: *the 2/3-loglog note is sharp for its architecture* [ET, Lemma 4.4], [PROVED]. Vaughan's argument and that of Pomerance–Weingartner [PW] are the case $\mathcal{K} = \{1\}$, giving $(\log N)^{2/3}$. For the 3/4 note, $h \asymp \kappa t$ and the same computation gives $\lambda^{3/4}$.

9. WHAT THE THEOREMS EXCLUDE

The following list is checked against the proofs of Theorems 4.5, 4.9, 7.6 and Lemma 3.4 [ET, §6.1]. Anything below lies outside the caps of Section 6.

- (1) *Majorants not ≥ 0 on all of $\mathbb{Z}$.* See Remark 3.6.
- (2) *Signed rounding.* Any bound exploiting cancellation in $\sum_{n \leq N} \nu(n) - N\mathbb{E}\nu$, or any rounding bound smaller than $\sum_i |a_i|$, for example one that uses which classes meet $[1, N]$.
- (3) *Non-CRT inputs.* The theorems see only the CRT law of residues. Type I/II sums, the arithmetic of $(p+a)/4$, Halász-type arguments, and majorants that are ≥ 1 only on exceptional *primes* and use prime equidistribution beyond the selector coordinates are outside. The theorems require $\nu \geq 1$ on the whole avoider set.
- (4) *Balanced and twin moduli.* Corollary 6.2 needs every modulus to have $P(M) \geq M^{1/(1+C)}$ with $C < 1$ fixed; Theorem 7.6 extends this to (η, B) -gapped $\mathcal{R}(M)$ -moduli with B fixed, and Theorem 7.7 to window-resolved families. Excluded are: families containing unresolved η -twin moduli with top prime in $(e^{\lambda^{1/4}}, e^{\lambda/2}]$; gapped moduli with $\log M / \log P(M)$ unbounded; and (a, D) - and Case-A moduli without a dominant prime. This part is not small (Lemma 9.1).
- (5) *Non-selector admissible sets.* The term $\log(Q_0/|R|)$ is uncontrolled (Remark 4.7). Corollary 6.2 handles the w_0 -smooth part only because $w_0 = O_C(1)$, and Theorem 7.6 because $W = W_0(B)$.
- (6) *Level and size.* Families with slice primes beyond $N^{O(1)}$ are formally outside, since Lemma 3.4 needs Λ_0 .
- (7) *Large multipliers.* In the prime-slice setting the small part q_0 of a modulus is free, but Corollary 6.1 needs $q_0 \leq \ell^C$. Conditions with $q_0 \gg \ell$, used

without paying their lcm in level, are not covered; no lower bound on that cost is proved [ET, §5.6(b)].

- (8) *Large sieve.* Only Montgomery's large sieve over a slice system, fibre by fibre, is covered.

Lemma 9.1 (balanced moduli carry cubic supply; [ET, Lemma 3.8]; [PROVED]). *There are $c > 0$ and x_0 such that for $x \geq x_0$,*

$$\sum_{M \leq x, M \equiv 3 \pmod{4}, P(M) \leq M^{1/2}} \frac{|\mathcal{R}(M)|}{M} \geq c(\log x)^3.$$

More precisely, for each fixed $\eta \in (0, 1/480]$, the moduli $M = k\ell_1\ell_2$ with primes $\ell_1 \in (Y, Y^{1+\eta/2}]$, $\ell_2 \in (Y^{1+\eta/2}, Y^{1+\eta}]$, odd $k \in (Y^{2\eta}, Y^{3\eta}]$ and $Y = x^{1/(2+4.5\eta)}$ already contribute $\geq c(\eta)(\log x)^3$; these are same-scale pairs, $\log \ell_2 / \log \ell_1 \leq 1 + \eta$.

Proof sketch. Each such M is $\leq x$ and balanced, and determines (k, ℓ_1, ℓ_2) . For coprime $u, v \leq z = Y^{1/8}$ with $(uv, k) = 1$ and $4uv \mid M+1$, the class $-uv^{-1} \pmod{M}$ lies in $\mathcal{R}(M)$, and distinct (u, v) give distinct classes modulo ℓ_2 . The ℓ_2 -sum over the progression $-(k\ell_1)^{-1} \pmod{4uv}$ is evaluated by partial summation, with the error terms controlled on average by Cauchy–Schwarz against Brun–Titchmarsh and the Bombieri–Vinogradov theorem [D, Ch. 28]. The (u, v) -sum is bounded below by the lattice lemma of [LL, Lemma 3.1], which needs $\eta \leq 1/480$. The k - and ℓ_1 -sums give $\gg \eta \log Y$ and $\gg \eta$. Full details are in [ET, §3.8], checked in [ETr, item 4]. $\square$

The share of the weighted Case-B mass carried by balanced moduli is 0.098, 0.121, 0.141, 0.157 and 0.171 at $x = 10^3, \dots, 10^7$, still growing [EVIDENCE]. Lemma 9.1 does not by itself show that every window partition loses: singleton windows have no internal pairs. So the exclusion of twin moduli is not cosmetic, but whether they can actually beat the Rankin functional is open.

10. SELBERG-TYPE MAJORANTS

10.1. A sieve limit with no structural hypothesis. Let $\Omega = \prod_i \Omega_i$ be a finite product probability space (CRT coordinates), with costs $s_i \geq 0$: $s_i = \log \ell$ for a slice prime and $s_i = 0$ for free coordinates. Let $\mathcal{A} \subseteq \Omega$ be *arbitrary*. Let V_κ be the span of functions of ω_T over sets T with $\sum_{i \in T} s_i \leq \kappa$; it is the sum of the Efron–Stein components H_T over these T .

Theorem 10.1 (Selberg-type sieve limit; [ET, Thm 5.5]; [PROVED]). *Let $g \in V_{\lambda/2}$ with $g \geq 1$ on $\mathcal{A}$, so that $\nu = g^2$ is a majorant of level λ . Then for every $\alpha > 0$,*

$$\mathbb{E}g^2 \geq e^{-\alpha\lambda/2} \frac{\mathbb{P}(\mathcal{A})^2}{\mathbb{P}(\omega \in \mathcal{A}, \omega' \in \mathcal{A})},$$

where ω' is the ρ -correlated copy: independently for each i , $\omega'_i = \omega_i$ with probability $\rho_i = e^{-\alpha s_i}$ and otherwise a fresh sample. Equivalently, $\log(1/\mathbb{E}\nu) \leq \alpha\lambda/2 + \Xi_{\mathcal{A}}(\alpha)$ with the noise-stability excess $\Xi_{\mathcal{A}}(\alpha) = \log[\mathbb{P}(\mathcal{A} \cap \mathcal{A}')/\mathbb{P}(\mathcal{A})^2] \in [0, \log(1/\mathbb{P}(\mathcal{A}))]$.

Proof. $\mathbb{E}[g\mathbf{1}_{\mathcal{A}}] \geq \mathbb{P}(\mathcal{A})$ and, since $g \in V = V_{\lambda/2}$, $\mathbb{E}[g\mathbf{1}_{\mathcal{A}}] = \langle g, \Pi_V \mathbf{1}_{\mathcal{A}} \rangle \leq \|g\| \|\Pi_V \mathbf{1}_{\mathcal{A}}\|$. Now

$$\begin{aligned} \|\Pi_V \mathbf{1}_{\mathcal{A}}\|^2 &= \sum_{c(T) \leq \lambda/2} \|(\mathbf{1}_{\mathcal{A}})_T\|^2 \leq e^{\alpha\lambda/2} \sum_T e^{-\alpha c(T)} \|(\mathbf{1}_{\mathcal{A}})_T\|^2 \\ &= e^{\alpha\lambda/2} \langle \mathbf{1}_{\mathcal{A}}, T_\rho \mathbf{1}_{\mathcal{A}} \rangle = e^{\alpha\lambda/2} \mathbb{P}(\mathcal{A} \cap \mathcal{A}'), \end{aligned}$$

where $c(T) = \sum_{i \in T} s_i$ and the noise operator T_ρ acts on H_T by $\prod_{i \in T} \rho_i$. $\square$

This covers Selberg's Λ^2 sieve. For a prime-slice system the excess factorises fibrewise, $\Xi_{\mathcal{A}_c}(\alpha) = \sum_\ell \log(1 + \rho_\ell p_\ell(c)/(1 - p_\ell(c))) \leq \frac{4}{3} \sum_\ell p_\ell(c) \ell^{-\alpha}$ when $p_\ell(c) \leq 1/4$, so Theorem 10.1 reproduces the Rankin functional [ET, Cor. 5.6]. Its point is that balanced, twin and shared-prime moduli are allowed: for Λ^2 -majorants, beating $3/4$ with balanced moduli requires the following hypothesis to fail.

Hypothesis 10.2 ($H_{\text{MS}}^{\text{Sel}}$; [ET, §5.7]; [OPEN]). For the complete Case-B and Case-A forced-class system with moduli $\leq e^\lambda$, with primes $\leq \lambda^4$ given $\rho = 1$, $\Xi_{\mathcal{A}}(\alpha) \ll \alpha^{-3}(\log \lambda)^{O(1)}$ uniformly for $\lambda^{-1} \leq \alpha \leq 1$.

Under Hypothesis 10.2, Theorem 10.1 caps Selberg-type majorants at $\lambda^{3/4}(\log \lambda)^{O(1)}$ for all moduli [CONDITIONAL]. A fibrewise version alone would not suffice: $\mathcal{A} = \{c = c_0\} \times \Omega'$ has $\Xi = 0$ in every nonempty fibre, yet $g = \mathbf{1}_{c=c_0}$ saves $\log Q_0$. The remedy is to tilt the fibre law.

Lemma 10.3 (fibre tilting; [TW, Lemma 6.6], [TW2, Lemma 2.1]; [PROVED]). *In the setting of Theorem 10.1, let F be a set of coordinates whose residues c (the fibre) range over $\mathbb{Z}/Q_F$; put $\mathcal{A}_c = \mathcal{A} \cap \{\text{fibre} = c\}$ and $\Xi_c(\alpha) = \log[\mathbb{P}(\mathcal{A}_c \cap \mathcal{A}'_c \mid c)/\mathbb{P}(\mathcal{A}_c \mid c)^2]$, where in $\mathcal{A}'_c$ the coordinates outside F are ρ -correlated and the fibre is shared. For any probability law P on fibres with $\mathcal{A}_c \neq \emptyset$ on its support, every $g \in V_{\lambda/2}$ with $g \geq 1$ on $\mathcal{A}$ satisfies*

$$\log \frac{1}{\mathbb{E}g^2} \leq \frac{\alpha\lambda}{2} + \log \left\| \frac{dP}{dU} \right\|_\infty + \mathbb{E}_P \Xi_c(\alpha).$$

For P uniform on a set R the middle term is $\log(Q_F/|R|)$.

Proof for uniform P on R . Let $\tilde{\sigma} = \sum_{c \in R} \pi_c U(\cdot \mid c, \mathcal{A})$ and $h = d\tilde{\sigma}/dU$. Then $\mathbb{E}_U[gh] \geq 1$, so $\mathbb{E}g^2 \geq 1/\|\Pi_{V_{\lambda/2}} h\|^2$, and $\|\Pi_V h\|^2 \leq e^{\alpha\lambda/2} \langle h, T_\rho h \rangle$. Since $\langle h, T_\rho h \rangle$ is nondecreasing in each ρ_i , put $\rho_i = 1$ on F ; then $\langle h, T_\rho h \rangle = Q_F \sum_{c \in R} \pi_c^2 e^{\Xi_c}$. Choosing $\pi_c \propto e^{-\Xi_c}$ and Jensen give $\leq (Q_F/|R|) e^{\text{avg}_R \Xi_c}$. The general case is [TW2, Lemma 2.1]. $\square$

10.2. Two large primes, twins included. *Setting* [TW2, §3]. Fix $B \geq 0$, $A_0 \geq 1$; let X be large, $L = \log X$, $W_1 = L^{1/2}$, $w_2 = L^8$ and $\alpha = L^{-1/4}$. The family is any set of classes $-4D \pmod{M}$ of $\mathcal{R}(M)$ with $M \leq X$, $M \leq P(M)^{1+B}$, and at most two distinct prime factors of M above w_2 . Twin moduli $k\ell_1\ell_2$ ($\ell_i > w_2$, k w_2 -smooth), dominant moduli and prime powers are included. All primes are charged, and the level is $\lambda \leq A_0 L$. The fibre is $c = n \pmod{w_2}$ the w_2 -smooth part, with a law P built from the QR base at primes $\leq W_1$, conditioning on avoidance of the small classes, and a good-fibre event; then $\log \|dP/dU\|_\infty \leq 4L^{1/2}$ [TW2, Lemmas 3.1–3.4]. In a fibre, a class $k\ell^v$ is unary at ℓ and a class $k\ell^v m^u$ is a binary edge.

The probabilistic core is a binary noise-stability bound. Take independent coordinates y_ℓ with laws ν_ℓ and a finite set of edges $e = \{(\ell, a), (m, c)\}$, $\ell \neq m$, hit when $y_\ell = a$ and $y_m = c$. Put $\pi_e = \nu_\ell(a)\nu_m(c)$, $\deg(\ell, a) = \sum \nu_m(c)$ over the edges at (ℓ, a) , $w_\ell = \sum_{e \ni \ell} \pi_e$ and $q_\ell = \sum_a \nu_\ell(a) \deg(\ell, a)^2$, and assume $w_\ell \leq \delta \leq 1/16$ for all ℓ . Let Z_1 be the probability that no edge is hit, and $Z_2(\bar{\rho})$ the probability that no edge is hit in either copy of the coupled pair in which, independently for each ℓ , $y'_\ell = y_\ell$ with probability $\bar{\rho}_\ell$ and is otherwise an independent copy.

Theorem 10.4 (binary noise stability; [TW2, Thm 1.4]; [PROVED]). *For every $\tilde{\rho} \in [0, 1]^{index\ set}$,*

$$\log \frac{Z_2(\tilde{\rho})}{Z_1^2} \leq (1 + 25\delta) \left[\sum_{e=\{\ell, m\}} \tilde{\rho}_\ell \tilde{\rho}_m \pi_e + \sum_j \tilde{\rho}_j q_j \right].$$

The proof [TW2, §1] differentiates $\log Z_2$ along $\kappa = t\tilde{\rho}$, uses an identity for $\partial_{\kappa_j} \log Z_2$ under the unconditioned law of the other coordinates, and bounds the result with a conditional form of the Lovász local lemma; no cluster expansion is needed. An exact check on 1200 small systems found no violation [EVIDENCE].

For a fibre c , a prime $j > w_2$ and a residue a , let $\deg_c(j, a)$ be the partner mass of the active binary classes through (j, a) , and $S_j(c) = \sum_a \nu_j(a) \min(\deg_c(j, a), 1)^2$.

Theorem 10.5 (two-prime Λ^2 cap, reduced; [TW2, Thm 5.1]; [PROVED]). *In this setting, for L large, every $g \in V_{\lambda/2}$ with $g \geq 1$ on the avoiders of the family satisfies*

$$\log \frac{1}{\mathbb{E}g^2} \leq \frac{A_0}{2} L^{3/4} + C_B L^{3/4} (\log L)^C + 11 \mathbb{E}_P \sum_{j > w_2} \rho_j S_j(c), \quad \rho_j = j^{-\alpha},$$

with an absolute exponent C .

The proof combines Lemma 10.3 with the fibre law P , Theorem 10.4 in each good fibre after a free quarantine of hub residues [TW2, Lemma 2.2], and a profile bound $\ll \alpha^{-3} (\log L)^{O(1)}$ for the unary and diagonal binary terms [TW2, Lemma 4.1].

The off-diagonal term splits by the labels of the classes. By [TW2, Lemma 5.3], the residue of $-4D \pmod{M}$ at a prime $j \mid M$ is $-u'/v'$, where $D/A = u'/v'$ in lowest terms; each class has the three *candidate rationals* $4D$, D/A , $1/(4\bar{D})$ with $\bar{D} = A^2/D$.

Lemma 10.6 (same-label part $(H_O^\pm)$; [TW2, Lemma 5.4]; [PROVED] mod the stated Brun–Titchmarsh and Shiu inputs). *The contribution to $\mathbb{E}_P \sum_j \rho_j S_j$ of pairs of classes through j that share a candidate rational, together with all prime-power classes, is $\ll_B (\log L)^{O(1)}$.*

Hypothesis 10.7 ($(H_O^\neq)$; [TW2, §5.1]; [OPEN]). Consider the pairs of classes through j that agree modulo j and have disjoint sets of candidate rationals. Their contribution to $\mathbb{E}_P \sum_{j > w_2} \rho_j S_j(c)$ is $\ll_B \alpha^{-3} (\log L)^{O(1)}$.

Corollary 10.8 ([TW2, Cor. 5.2]; [CONDITIONAL] on $(H_O^\neq)$). *In this setting, $\log(1/\mathbb{E}g^2) \ll_{A_0, B} L^{3/4} (\log L)^{O(1)}$: the Λ^2 cap for $\mathcal{R}(M)$ -families with at most two prime factors above $(\log X)^8$, twin moduli included.*

Proof. Theorem 10.5, Lemma 10.6 and Hypothesis 10.7. □

For the pairs in $(H_O^\neq)$, every difference of candidate rationals is a nonzero integer divisible by j . The random model predicts a contribution $\ll L^{6+o(1)}/w_2 = o(1)$, against a trivial bound $\asymp L^6/\alpha$; numerically the relevant counts are at or below the random prediction in all six tested rows [EVIDENCE]. The restriction $M \leq P(M)^{1+B}$ with B fixed is genuine: for fixed B , twins $k\ell_1\ell_2$ with a w_2 -smooth cofactor $k > M^{(B-1)/(B+1)}$ are excluded.

11. OPEN PROBLEMS

A proof of $\theta > 3/4$ by a congruence sieve must therefore contain at least one of: (i) a joint use of unresolved η -twin (or other balanced) moduli that beats their mass; (ii) multipliers larger than the slice primes, used without paying their lcm in level; (iii) a non-selector admissible set with a large saving of its own; (iv) a non-CRT input, or signed cancellation in the rounding errors [ET, §6]. The following statements would close (i) and (iii).

Hypothesis 11.1 (H_{MS} ; [ET, §5.6]; [OPEN]). For every CRT system of forced classes with moduli $\leq e^\lambda$, every majorant of level λ satisfies $\log(1/\mathbb{E}\nu) \leq C \inf_\alpha [\alpha\lambda + \sum_{\text{cond}} \mathbb{P}(\text{cond})(\text{mod cond})^{-\alpha}] + O(\log^2 \lambda)$.

Under H_{MS} every CRT-majorant architecture built from Case-B and Case-A forced classes, with a bounded-saving admissible set and rounding treated in absolute value, is capped at $(\log N)^{3/4}$ [CONDITIONAL]. H_{MS} is open even for one scale of primes with random pair conditions.

For the twin residual, the reduction of [TW, §6] isolates an arithmetic-free statement. Let V be a block of primes with costs $s_\ell \in (s, (1+\eta)s]$ and $d = \lfloor \lambda/s \rfloor$.

Conjecture 11.2 (k -ary comparison inequality; [TW, Conj. 6.4]; [OPEN]). Let $\nu = \bigotimes \nu_\ell$ be a product law on V , and let $\mathcal{F}$ be a family of hard k -ary constraints ($2 \leq k \leq r$) on V with all incident weights $\leq \theta \leq \theta_0(r)$ and total ν -mass $\mu_{\geq 2}$. Let σ be the sequential law: in increasing order of ℓ , ν_ℓ conditioned off the residues activated by earlier coordinates, or ν_ℓ itself if that set has ν_ℓ -mass $> \delta_\ell$. Then for every λ -level $f \geq 0$,

$$\mathbb{E}_\sigma f \leq \exp(C_r[d \log(2 + \mu_{\geq 2}) + 1]) \mathbb{E}_\nu f.$$

Here the *incident weight* of a residue a at ℓ is the ν -probability that some activated constraint through (ℓ, a) is completed by the other coordinates.

Proposition 11.3 ([TW, Prop. 6.5]; [CONDITIONAL] on Conjecture 11.2 and Hypothesis K2). Assume Conjecture 11.2 for $r = \lfloor (1+B)(1+\eta) \rfloor$ and the k -ary second-moment hypothesis K2 of [TW, §6.3]. Then for every fixed B , every family of $\mathcal{R}(M)$ -classes with $M \leq P(M)^{1+B}$ (twin, prime-power top and all other shapes included), plus W_0 -smooth classes, has saving $\ll_B \eta^{-1} \lambda^{3/4} \log \lambda$.

The extra $\log \lambda$ is the price of the weak form; the evidence for Conjecture 11.2 is toy-scale exact linear programmes only [EVIDENCE].

For Λ^2 -majorants the core is a sparse noise-stability statement.

Conjecture 11.4 (sparse noise stability; [TW, Conj. 6.8]; [OPEN]). Let X_i be independent and $\mathcal{E}$ a finite family of events, each depending on coordinates $S(E)$, with incident weights $\sum_{E \ni i} \mathbb{P}(E) e^{a|S(E)|} \leq \theta \leq \theta_0(a)$ and a codegree condition for $|S(E)| \geq 3$. Let $\mathcal{A}$ be “no event” and $\mathcal{A}'$ its ρ -correlated copy. Then

$$\log \frac{\mathbb{P}(\mathcal{A} \cap \mathcal{A}')}{\mathbb{P}(\mathcal{A})^2} \leq (1 + O(\theta)) \sum_E \mathbb{P}(E) \left(\prod_{i \in S(E)} (\rho_i + (1 - \rho_i) P_i(E)) - \mathbb{P}(E) \right),$$

where $P_i(E)$ is the marginal factor of E at i .

Theorem 10.4 proves a bound of this type for binary systems ($|S(E)| \leq 2$) with the diagonal and one-site terms on the right, and Theorem 10.5 reduces the two-prime Λ^2 cap to the arithmetic Hypothesis 10.7. A cleaner sufficient form of the

latter is a second-moment equidistribution statement for divisor labels of $A \equiv 4^{-1} \pmod{j}$ modulo j , averaged over j [TW2, §5.5, H_{div}] [OPEN]. Known results on divisor functions in progressions do not reach it: it is a second moment with two coupled congruences at moduli up to X .

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